

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 | SUB-STRAND 1.3 TEMPERATURE AND THERMAL EXPANSION | FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why did the sealed water bottle crack in the freezer, and what does this reveal about how all matter responds to changes in temperature? In this Final Explanation you tell the complete story of the cracked frozen bottle. A learner filled a clean 500 ml glass bottle to the brim with water, sealed it tightly with a metal cap and left it in a freezer overnight. By morning the bottle was cracked and a column of ice stuck out where the cap had been. Answer all five sections in the space under each prompt. Refer back to the cracked bottle throughout. Use scientific words accurately (temperature, kinetic energy, particles, thermal expansion, expansivity, density, lattice). If you can, you may also use the term hydrogen bonding, which you met in Lesson 6. Show every step of any calculation, with units. Draw and label clear particle diagrams where asked. Use examples from Kenya wherever you can. Write about 1,200 to 1,800 words in total. Write in your own words and use evidence from the experiments and graphs you worked with in Lessons 1 to 6.

Section 1: Explaining the Crack at the Particle Level

Look back at your Lesson 1 prediction and baseline model of the cracked bottle. (a) Draw and label two particle diagrams: liquid water just before freezing, and the ice formed in the bottle. (b) Using your diagrams, explain why the same mass of water takes up a different volume as ice than as liquid water. (c) Explain, step by step, how this led to the bottle cracking. Include the words sealed, pressure and tensile strength. (d) Most everyday materials get smaller when cooled. Explain why the cracked bottle surprised you in Lesson 1.

Write your answer here:

Section 2: Temperature and How We Measure It

(a) Explain what temperature tells us about the particles in a substance. **(b)** Water in the bottle began at about 18 °C and the freezer was at -18 °C. Convert both temperatures, and the freezing point of water, to kelvin. State the conversion you used. **(c)** Describe how a liquid-in-glass thermometer works and name its thermometric property. Then explain why the clinical thermometer you used at Station B would not be a suitable choice for measuring the freezer temperature. **(d)** Explain what happens to the kinetic energy of the water molecules as the freezer cools them, and link this to the cracked bottle.

Write your answer here:

Section 3: Expansion of Solids and Quantifying It

- (a) Explain at the particle level why a solid metal rod gets longer when heated. (b) Write the formula for linear expansion and say what each symbol means. (c) On the Standard Gauge Railway between Mombasa and Nairobi, a steel rail is 25 m long and its temperature rises by 45 °C. Using $\alpha = 11 \times 10^{-6}$ per °C, calculate the increase in length in millimetres. (d) A 10 m steel beam on a bridge in Kisumu experiences a 15 °C rise. Calculate its expansion in millimetres. (e) Explain why engineers leave gaps, and how this idea links to the cracked bottle.

Write your answer here:

Section 4: Water Is Different: Evidence and Calculation

In Lesson 4 you cooled water in a flask with a capillary tube and plotted a temperature-volume graph. (a) Describe the shape of your graph as the water cooled from 20 °C to 0 °C and say what it shows about the volume at about 4 °C. (b) State the temperature at which water has its greatest density, and say how this differs from most substances. (c) Liquid water has a density of about 1.00 g/cm³ and ice at 0 °C about 0.917 g/cm³. Taking 500 ml of liquid water to have a mass of 500 g, calculate the volume of the ice and the increase in volume in ml, and give this as a percentage of 500 ml. (d) Explain why the formula $\Delta V = \gamma V_0 \Delta T$ does not by itself describe this freezing change.

Write your answer here:

Section 5: Using Our Understanding in Kenya

(a) Describe one Kenyan example from Lesson 5 where thermal expansion is a problem engineers must design for, and one where it is used deliberately as a helpful tool. Name the expansion principle at work in each. (b) Kenya Power copper cables run between Nairobi and Thika. Explain why the cables are installed with some slack, and say what would happen on a cold morning if they were installed pulled tight. (c) Water pipes in cold highland areas such as Kericho and Nyandarua can burst. Use your answers above to explain why. (d) Return to the driving question. In your own words, what does the cracked bottle reveal about how matter responds to temperature changes?

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Particle-level explanation of the cracked bottle	Clear labelled diagrams of liquid water and the hexagonal ice lattice. Explains greater spacing, larger volume, sealed container, outward pressure and tensile strength in a correct logical chain.	Explains that ice has more spacing and takes up more volume, and that the sealed bottle cracked because of pressure, but the chain of reasoning or the diagrams have small gaps.	States that water expands or that the bottle cracked but gives little or no particle-level reasoning, or diagrams are missing or incorrect.
Temperature, scales and thermometers	Defines temperature as average kinetic energy, converts all temperatures to kelvin correctly (291 K, 255 K, 273 K), and explains thermometric property and thermometer types accurately.	Mostly correct definition and conversions with a minor error or incomplete explanation of how thermometers work.	Definition of temperature is incorrect or confused with heat, or conversions and thermometer descriptions are mostly wrong or missing.
Calculations and use of evidence	All calculations (12.4 mm, 1.65 mm, 545 ml, 45 ml, 9%) are set out with formula, substitution, correct units and answers, and the temperature-volume graph is accurately described as evidence.	Most calculations are correct with steps shown; one slip in units or arithmetic, or the graph description is partly complete.	Calculations are incomplete or incorrect, steps are not shown, or evidence from the graph is not used.
Anomalous behaviour of water	Accurately describes the U-shaped graph, maximum density at 4 °C, the change in density from 1.00 to 0.917 g/cm ³ , and explains why $\Delta V = \gamma V_0 \Delta T$	Describes the anomaly and 4 °C correctly but gives limited explanation of density change or of the formula's limits.	Does not identify 4 °C or the anomaly clearly, or treats water like all other substances.

	does not by itself describe freezing.		
Kenyan applications and communication	Connects the bottle to at least two Kenyan examples (SGR, Thika Superhighway, Kenya Power cables, burst pipes in Kericho or Nyandarua) with correct principles, and answers the driving question clearly using accurate scientific vocabulary in organised writing.	Uses one or two Kenyan examples correctly, with a reasonable link to the driving question and mostly accurate vocabulary.	Examples are missing, vague or incorrect, the link to the cracked bottle is weak, and the writing is hard to follow.

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Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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